

NIHAL MALAVALLI LOKESH

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Summary

Data science graduate student experienced in developing and evaluating ML models across real-world datasets, with strong skills in Python, feature engineering and performance analysis. Built projects spanning classical ML and deep learning (including RL), with emphasis on measurable results, clean pipelines and visual reporting.

EXPERIENCE

University of Texas at Arlington

Research Assistant

Arlington · Aug 2025 - May 2026

- Engineered an intelligent classification pipeline combining unsupervised learning (K-means clustering) with domain knowledge integration to develop an autonomous rock-typing agent that classifies carbonate reservoirs into 8 hydraulic flow units, achieving 69% error reduction through iterative model optimization and feature-driven decision logic.
- Developed a data orchestration pipeline (472+ samples, multi-source integration) with intelligent feature transformation (RQI, FZI, T2 geometric mean), implementing automated quality validation and anomaly detection to prepare high-quality training data for downstream ML models.
- Conducted statistical analysis using OLS regression with bootstrap resampling (R=10,000) to quantify model uncertainties and validate predictive performance across train/test splits, using Python (NumPy, Pandas, Scikit-learn, Statsmodels).

Student Administrative Assistant

Arlington · Aug 2025 - May 2026

- Built and maintained **Power Automate workflows** to extract event data from Mazevo and populate standardized **Excel operational sheets**.
- Reduced manual data entry time by **95%** by automating Mazevo to Excel reporting using Power Automate + Office Scripts.
- Processed **90+ bookings/week** and ensured accuracy across schedules, setups and service requirements with minimal errors.

VMX HI Connectors

Design Engineering Intern

Aug 2022 - Nov 2022

- Led optimization of connector design, improving wear efficiency through analytical evaluation and iterative redesign.
- Assisted in data collection, statistical analysis and reporting to streamline manufacturing processes.
- Gained hands on exposure to electroplating workflows (e.g., Zinc-Nickel) and post-treatment concepts (e.g. embrittlement relief), tying coating choice to corrosion and conductivity needs.
- Documented process improvements contributing to a 10% increase in production efficiency.

EDUCATION

University of Texas at Arlington

Master's, Data Science

Aug 2024 - Present

Arlington

- GPA:** 3.8/4
- Coursework:** Machine Learning, Statistical Analysis, Data Mining, Data Visualization, Database Management Systems, Python Programming

Visvesvaraya Technological University

Bachelor's, Mechanical Engineering

Sep 2019 - May 2023

- GPA:** 3.5/4 CGPA

PROJECTS

Reinforcement Learning Experiment: Double DQN (ALE/Freeway-v5) <https://github.com/B1a-z3/Reinforcement-learning-Double-DQN>

- Designed and trained an autonomous decision-making agent using Deep Q-Network (DQN) architecture with learned exploration-exploitation policies, achieving progressive policy convergence over **300 episodes** with optimal performance validation (**22 reward** peak, sustained 18-20 convergence).
- Designed an exploration strategy (ϵ : **1.0** -> **0.1**, **decay 0.995**) and optimized learning stability with γ =**0.99**, **batch=16** and **Adam lr = 2.5e-5**.
- Produced both quantitative and qualitative evaluation outputs (reward plot + rollout clip) to demonstrate learning progress and policy behavior.

Ksat Prediction Using Machine Learning https://github.com/B1a-z3/KSAT_Quest_Regression_Runoff_NERDY_POTATOES

- Developed regression models (Random Forest, XGBoost, LightGBM) to predict saturated hydraulic conductivity (Ksat) based on soil properties.
- Performed extensive feature engineering, missing value handling and hyperparameter tuning to optimize performance.
- Achieved significant accuracy improvements using R score of 0.88 as evaluation metrics.

Classification of Psychiatric Disorders using EEG Signals https://github.com/B1a-z3/EEG_Analysis

- Designed and trained 2D CNNs integrated with ResNet blocks to classify psychiatric disorders from EEG recordings.
- Enhanced model performance using Principal Component Analysis (PCA) for dimensionality reduction and Synthetic Minority Over-sampling Technique (SMOTE) for class balancing.
- Visualized EEG feature maps and correlation matrices to extract clinically relevant patterns.

Data Visualization and EDA on Terrorism Dataset https://github.com/B1a-z3/Data_Viz

- Conducted exploratory data analysis on terror attack data across multiple countries and years.
- Created interactive dashboards using Plotly and Tableau, showcasing regional attack patterns, weapon type distributions and success & failure trends.
- Compared visualization tools (Seaborn, Matplotlib, ggplot2) based on output quality and usability for data analysis.

SKILLS

- Programming:** Python (Pandas, NumPy, Scikit-learn), SQL, HTML/CSS, Flask
- Data Science:** Machine Learning, Data Analysis, Data Visualization (Tableau, Matplotlib, Seaborn, Plotly, PowerBI), Feature engineering, Anomaly Detection
- Generative AI & Large Language Models:** LLM fundamentals, Prompt engineering, Fine-tuning & adaptation, In-context learning
- Tools & Frameworks:** Git/GitHub, AutoCAD, SolidWorks, Ansys Workbench
- Other Skills:** Statistical Analysis, Database Management, Object-Oriented Programming (OOP)

CERTIFICATIONS

- Google Data Analytics Professional Certificate Coursera
- SolidWorks Certified Professional
- Cybersecurity Fundamentals